

Se inicio en el primer documento el proyecto con tests nulos, sin cobertura.

Posteriormente se añadieron unos primeros tests que también se indicaron en el entregable 1, para realizar las pruebas iniciales.

En este segundo entregable nos centramos en las evidencias de la calidad del código y tests realizados, con cobertura mediante sonarQube cloud, asociado a nuestro repositorio github directamente.

Todo se ha realizado en la rama que completamos en el entregable anterior:

https://github.com/Jero-to/app_gastos/tree/feature/consultas-y-filtros

Los pasos a seguir han sido:

- Creación de ficheros de configuración pytest.ini y .coveragec (visibles en el repositorio)

 Jero and Jero test: add test suite for filtros and gastos with coverage 5c3951f · 7 minutes ago 5 Commits		
 .github/workflows	test: add test suite for filtros and gastos with coverage	7 minutes ago
 .hypothesis	test: add test suite for filtros and gastos with coverage	7 minutes ago
 .kiro/specs/gastos-filtros-consultas	feature: añadir consultas y filtros con ADR y especificacio...	1 hour ago
 .scannerwork	test: add test suite for filtros and gastos with coverage	7 minutes ago
 Entregables	test: add test suite for filtros and gastos with coverage	7 minutes ago
 docs	test: add test suite for filtros and gastos with coverage	7 minutes ago
 tests	test: add test suite for filtros and gastos with coverage	7 minutes ago
 .DS_Store	test: add test suite for filtros and gastos with coverage	7 minutes ago
 .coveragerc	test: add test suite for filtros and gastos with coverage	7 minutes ago
 .gitignore	creación del módulo principal y primeros elementos añadi...	2 hours ago
 README.md	Initial commit	3 hours ago
 filtros.py	feature: añadir consultas y filtros con ADR y especificacio...	1 hour ago
 gastos.py	feature: añadir consultas y filtros con ADR y especificacio...	1 hour ago
 pullRequest.png	test: add test suite for filtros and gastos with coverage	7 minutes ago
 pull_request_template.md	test: add test suite for filtros and gastos with coverage	7 minutes ago
 pytest.ini	test: add test suite for filtros and gastos with coverage	7 minutes ago
 sonar-project.properties	test: add test suite for filtros and gastos with coverage	7 minutes ago

- Generar nuevos tests que amplíen la cobertura hasta casi el total de código. Visibles en la carpeta “tests” del repositorio:

```
test_filtros.py x
tests > test_filtros.py

74 def test_total_por_categoria_de_un_mes(gastos):
78
79
80 def test_total_por_categoria_lista_vacia():
81     res = total_por_categoria([])
82     assert sum(res.values()) == 0
83
84
85 # -----
86 # filtrar_por_categoria
87 # -----
88
89 def test_filtrar_por_categoria(gastos):
90     res = filtrar_por_categoria(gastos, "Alimentación")
91     assert len(res) == 2
92     assert all(g["categoria"] == "Alimentación" for g in res)
93
94
95 def test_filtrar_por_categoria_sin_resultados(gastos):
96     assert filtrar_por_categoria(gastos, "Viajes") == []
97
98
99 def test_filtrar_lista_vacia():
100     assert filtrar_por_categoria([], "Ocio") == []
101
102
103 # -----
104 # ordenar_por_importe
105 # -----
106
107 def test_ordenar_descendente_por_defecto(gastos):
108     res = ordenar_por_importe(gastos)
109     assert [g["cantidad"] for g in res] == [50.25, 45.00, 30.00, 20.10, 12.50]
110
111
112 def test_ordenar_no_modifica_la_lista_original(gastos):
113     copia = list(gastos)
114     ordenar_por_importe(gastos)
115     assert gastos == copia
116
117
118 def test_ordenar_lista_vacia():
119     assert ordenar_por_importe([]) == []
120
121
122 # -----
123 # Tests de propiedad (hypothesis)
124 # -----
125
```

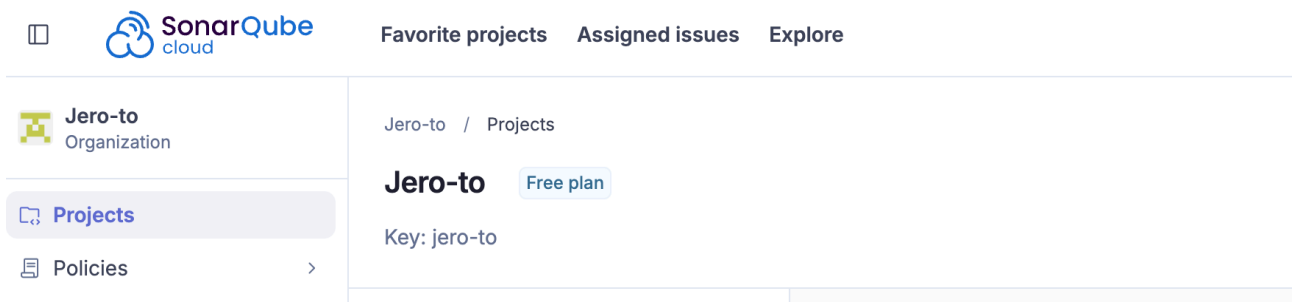
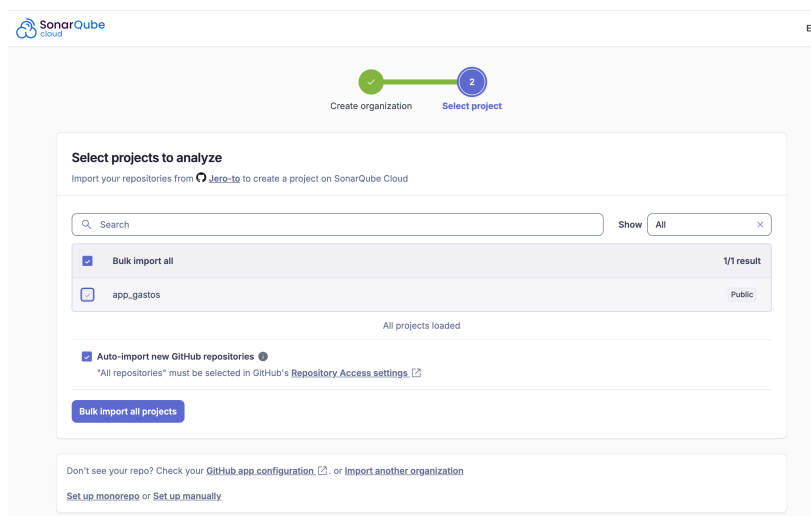
- Realizar mediciones para comprobar la cobertura local:

```
((.venv) jero@MacBook-Pro-de-Jero app_gastos % pytest --cov=filtros --cov=gastos --cov-branch --cov-report=term-missing -j
-cov-report=html --cov-report=xml
===== test session starts =====
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/jero/Cursos/gastos/app_gastos
configfile: pytest.ini
testpaths: tests
plugins: cov-7.1.0, hypothesis-6.141.1
collected 30 items

tests/test_filtros.py ..... [ 66%]
tests/test_gastos.py ..... [100%]

===== tests coverage =====
----- coverage: platform darwin, python 3.9.6-final-0 -----
Name          Stmts  Miss Branch BrPart  Cover   Missing
-----
filtros.py      24      2      4      0    93%    22-23
gastos.py     149     64     50      5    57%    59-60, 93-95, 150-166, 172-198, 204-213, 219-227, 244
-----
TOTAL          173     66     54      5    62%
Coverage HTML written to dir htmlcov
Coverage XML written to file coverage.xml
===== 30 passed in 2.01s =====
```

Con esto, iniciar la configuración y carga del proyecto en SonarCloud:



<https://sonarcloud.io/organizations/jero-to/projects>

Posteriormente se ha realizado la ejecución de sonar-scanner y se ha obtenido una cobertura inicial de un 61% aproximadamente:

```
app_gastos -- -zsh -- 120x61
23:29:04.310 INFO Sensor SonarPython A3S Context Collector [python]
23:29:04.310 INFO Sensor SonarPython A3S Context Collector [python] (done) | time=0ms
23:29:04.310 INFO Sensor pySecurityContextCollectorSensor [security]
23:29:04.310 INFO Sensor pySecurityContextCollectorSensor [security] (done) | time=0ms
23:29:04.310 INFO Sensor Scan Manifest A3S Context Collector [commona3scontextcollector]
23:29:04.310 INFO Sensor Scan Manifest A3S Context Collector [commona3scontextcollector] (done) | time=0ms
23:29:04.310 INFO Sensor CveReachabilitySensor [cvereachability]
23:29:04.310 INFO Skipping CveReachabilitySensor
23:29:04.310 INFO Sensor CveReachabilitySensor [cvereachability] (done) | time=0ms
23:29:04.310 INFO Sensor IaC Project Sensor [iac]
23:29:04.313 INFO Sensor IaC Project Sensor [iac] (done) | time=3ms
23:29:04.315 INFO Sensor PythonArchitectureSensor [architecture]
23:29:04.321 INFO Found 1 potential Udg file location(s) for "py" in "/Users/jero/Cursos/gastos/app_gastos/.scannerwork
"
23:29:04.321 INFO - /Users/jero/Cursos/gastos/app_gastos/.scannerwork/architecture/py
23:29:04.328 INFO Architecture analysis is enabled.
23:29:04.332 INFO * Protobuf reading starting | memory total=880 | free=253 | used=626 (MB)
23:29:04.333 INFO * Reading SonarArchitecture UDG data from directory "/Users/jero/Cursos/gastos/app_gastos/.scannerwor
k/architecture/py"
23:29:04.362 INFO * Files successfully loaded: "2" out of "2"
23:29:04.363 INFO * Purging externals (components not scanned) from graphs
23:29:04.363 INFO * Purging excluded nodes from graphs
23:29:04.364 INFO * Protobuf reading complete | memory total=880 | free=249 | used=630 (MB)
23:29:04.376 INFO * Build architecture.graph.py.file_graph.default_perspective hierarchy graph complete (filtered=false
) | memory total=880 | free=240 | used=639 (MB)
23:29:04.413 INFO No directives configured
23:29:04.424 INFO * No intended architecture defined or detected, using an empty model
23:29:04.498 INFO Boundary detection is disabled.
23:29:04.500 INFO Sensor PythonArchitectureSensor [architecture] (done) | time=185ms
23:29:04.500 INFO Sensor Zero Coverage Sensor
23:29:04.500 INFO Sensor Zero Coverage Sensor (done) | time=0ms
23:29:04.500 INFO Sensor Architecture Telemetry [architecture]
23:29:04.501 INFO Sensor Architecture Telemetry [architecture] (done) | time=0ms
23:29:04.524 INFO ----- Gather SCA dependencies on project
23:29:04.866 INFO Dependency analysis skipped
23:29:04.872 INFO CPD Executor Calculating CPD for 2 files
23:29:04.890 INFO CPD Executor CPD calculation finished (done) | time=18ms
23:29:04.892 INFO SCM Publisher SCM provider for this project is: git
23:29:04.893 INFO SCM Publisher 4 source files to be analyzed
23:29:05.021 INFO SCM Publisher 2/4 source files have been analyzed (done) | time=128ms
23:29:05.022 WARN Missing blame information for the following files:
23:29:05.022 WARN * tests/test_gastos.py
23:29:05.022 WARN * tests/test_filtros.py
23:29:05.023 WARN This may lead to missing/broken features in SonarCloud
23:29:05.028 INFO SCM revision ID 'f20ceab0c9c9e4b59b240b41bd17502cfc6c8c6b'
23:29:05.231 INFO Analysis report generated in 207ms, dir size=256 KB
23:29:05.246 INFO Analysis report compressed in 14ms, zip size=44 KB
23:29:05.482 INFO Analysis report uploaded in 236ms
23:29:05.490 INFO ANALYSIS SUCCESSFUL, you can find the results at: https://sonarcloud.io/dashboard?id=jero-to_app_gast
os
23:29:05.490 INFO Note that you will be able to access the updated dashboard once the server has processed the submitte
d analysis report
23:29:05.490 INFO More about the report processing at https://sonarcloud.io/api/ce/task?id=AaDvEj6AGrBLBxcDFDcs
23:29:05.491 INFO ----- Upload SCA dependency files
23:29:06.843 INFO Successfully sent architecture data
23:29:07.600 INFO Sensor cache published successfully
23:29:07.777 INFO Analysis total time: 25.267 s
23:29:07.784 INFO SonarScanner Engine completed successfully
23:29:08.289 INFO EXECUTION SUCCESS
23:29:08.292 INFO Total time: 31.040s
(.venv) jero@MacBook-Pro-de-Jero app_gastos %
```

★ **jero-to_app_gastos** New Private

Last analysis: 29 Sept 2026, 23:28 • 193 Lines of Code • Python

0	0	0	100%	61.8%	0.0%
Security	Reliability	Maintainability	Hotspots Reviewed	Coverage	Duplications

Para mejorar este resultado, mediante un prompt sencillo se ha pedido a Kiro que consiga ampliar la cobertura de tests y el resultado ha sido óptimo:

98% sobre las fuentes. La única línea sin cubrir es la 244 (`if __name__ == "__main__"`), que Sonar suele excluir automáticamente o tiene impacto mínimo.

Para regenerar el `coverage.xml` que sube a Sonar:

```
bash
.venv/bin/pytest --cov=filtros --cov=gastos --cov-
```

Resumen de lo añadido en `test_gastos.py`:

- Tests para opciones 4, 5, 6 y 7 (total mensual, total por categoría, filtrar, ordenar)
- Ramas de error: año/mes no numérico, mes fuera de rango, categoría inválida
- Casos vacíos (sin gastos) para las opciones que lo soportan
- `test_anadir_cantidad_invalida` y `test_anadir_gasto_categoria_invalida` con selección no numérica

Est. Credits Used: 2,04 Elapsed time: 1m 11s

```
(.venv) jero@MacBook-Pro-de-Jero app_gastos % .venv/bin/pytest --cov=filtros --cov=gastos --cov-report=xml
```

```
===== test session starts =====
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/jero/Cursos/gastos/app_gastos
configfile: pytest.ini
testpaths: tests
plugins: cov-7.1.0, hypothesis-6.141.1
collected 48 items

tests/test_filtros.py ..... [ 41%]
tests/test_gastos.py ..... [100%]

===== tests coverage =====
coverage: platform darwin, python 3.9.6-final-0

Coverage XML written to file coverage.xml

===== 48 passed in 2.03s =====
(.venv) jero@MacBook-Pro-de-Jero app_gastos %
```

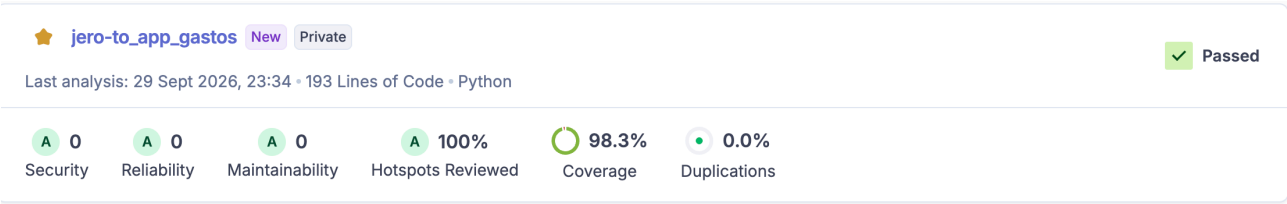
```
(.venv) jero@MacBook-Pro-de-Jero app_gastos % pytest --cov=filtros --cov=gastos --cov-branch --cov-report=term-missing -j
-cov-report=html --cov-report=xml
===== test session starts =====
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/jero/Cursos/gastos/app_gastos
configfile: pytest.ini
testpaths: tests
plugins: cov-7.1.0, hypothesis-6.141.1
collected 48 items

tests/test_filtros.py ..... [ 41%]
tests/test_gastos.py ..... [100%]

===== tests coverage =====
coverage: platform darwin, python 3.9.6-final-0

Name      Stmts  Miss Branch BrPart  Cover  Missing
-----
filtros.py   24     2     4      0   93%  22-23
gastos.py  149     1    50      1   99%  244
-----
TOTAL       173     3    54      1   98%
Coverage HTML written to dir htmlcov
Coverage XML written to file coverage.xml
===== 48 passed in 2.16s =====
```

Tras una nueva ejecución de sonar-scanner se ha obtenido el objetivo que se requería.



Así obtenemos el objetivo de este segundo entregable. Partimos de una cobertura de tests mínima, vamos ampliándola y rematamos con una verificación de calidad del proyecto gracias a SonarQube.